



Complex Systems

2020/2021

Exam N 18 of June 2021 Time: 2h00

Name:

Number:

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Pergunta 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use a concrete example to **help you** to explain the concept.

a) Fixed-Point.

Your answer:

b) Chaotic System.

Your answer:

c) Mandelbrot Set .

Your answer:

Pergunta 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) Non-autonomous, higher-order differential equations can always be converted into autonomous, first-order forms.

Your answer:

Your answer:

- b) One-dimensional differential **equation** equations are not capable of periodic or chaotic behavior.

Your answer:

Your answer:

Pergunta 3 - Simule 25 points

Consider a 1D cellular automata with neighborhood of radius 1. Admit that someone said that he/she observe the following state transition:

$$t : 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 0$$

$$t + 1 : 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0$$

Is this situation possible? Justify your answer. What was the transition rule used and, if erroneous, correct it and present the new result.

Your answer:

Your answer:

Pergunta 4 - Explain 25 points

The Sierpinski Triangle is an object of study in complex systems. In class, we discussed different ways of obtaining this object. Describe all of them with rigor and identify the conceptual ideas behind each approach, if any.

Your answer:

Your answer:

Pergunta 5 - Model 20 points

Consider the system described by the following difference equation:

$$x_{n+1} = \sqrt{x_n}$$

Identify the fixed points of the system, if any, describe their stability and state the long term behavior of the system. Draw the corresponding phase line.

Your answer:

Your answer:

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